Abstract

Cancer drug response is not often combined with personalized plans for patients who have specific cancer types and genomic profiles, leading to long-term treatment plans with less efficacy. Currently, genomic features are not considered triggering variability in cancer drug response. Therefore, multi-gene pharmacogenetic testing was examined to increase drug sensitivity from patients receiving treatment and prescribing recommendations based on genotype results. Shifting the focus to matching treatments to biological variables to reduce the size of various cancerous tumors and minimize side effects.

To address this problem, multiple machine learning models were developed to predict drug sensitivity (LN_IC50) using genomic and pharmacological features. After preprocessing and feature selection, several models were evaluated, with the final approach using a hybrid stacked model that combines the CatBoost and XGBoost predictions through a second-layer linear regression model to improve the LN_IC50 prediction. This approach allows the model to use the complementary strengths of both gradient boosting models rather than relying on a single hypertuned model's prediction. The results indicated that drug response can be predicted with relatively high accuracy, showing that genomic features provide meaningful insight into treatment effectiveness. This approach is not just designed to only generate predictions, but to support a clearer and more efficient way for medical professionals to evaluate and compare potential drug treatments for individual patients. By combining data analytics and cancer pharmacogenomics, this analysis tests the possibility of a decrease in timelines of personalized cancer treatment plans due to an increase in efficacy of the treatment type.